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Original Article

# Dynamic recrystallization behavior and microstructure evolution of low-density high-strength Fe–Mn–Al–C steel

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ABSTRACT

The hot deformation behavior of low-density high-strength Fe–Mn–Al–C steel at $T = 900$ –1150 °C and $\dot{\varepsilon} = 0.01$–10 $s^{-1}$ is studied using the Gleeble-3500 thermo-mechanical simulator. The rheological stress curve characteristics of the steel are analyzed through experimental data and the dynamic recrystallization activation energy is calculated. The critical strain/stress, peak strain/stress and steady-state strain/stress models of dynamic recrystallization are characterized by introducing the dimensionless parameter Zener-Hollomon. The microstructure evolution is analyzed and the dynamic recrystallization state diagram of the steel is drawn. The results indicated that the steel is a negative temperature-sensitive and a positive strain rate-sensitive material. The dynamic recrystallization activation energy is 669.844 kJ $mol^{-1}$. With the decrease of $\dot{\varepsilon}$ or the increase of $T$, the $Z$ parameter decreases gradually and the dynamic recrystallization behavior is more likely to occur; however, its grain size increases accordingly. The fine and uniform grain structure can be obtained by hot working in the range of 950–1000 °C/0.01–0.1 $s^{-1}$ and 1050 –1100 °C/1–10 $s^{-1}$. Finally, the prediction model for the dynamic recrystallization average grain size is established, and the correlation coefficient ($R$) of the prediction model is 0.996; therefore, the model has a high prediction accuracy.

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## 1. Introduction

With the rapid development of the automobile industry, automotive lightweight materials have become an important research topic. Among them, Fe–Mn–Al–C steel has good corrosion resistance, low density and excellent comprehensive mechanical properties, having great application prospect in the fields of weight reduction, collision safety and the complex working conditions of the automobiles structure [1–4]. In addition, thermoplastic deformation can effectively refine the grain size and improve the properties of forgings and microstructure. In the field of actual production, the microstructure is primarily controlled by way of thermal machining. Dynamic recrystallization (DRX) is beneficial to grain refinement and homogenization and is an important deformation mechanism to improve the microstructure and properties within the hot working process [5–7]. Therefore, in-depth studies on the microstructure evolution and DRX behavior of Fe–Mn–Al–C steel are of great significance for the formulation and optimization of the hot forging process of the steel.

At present, reports on Fe–Mn–Al–C steel primarily focus on smelting, heat treatment and mechanical properties. For example, Moon et al. [8] studied Fe–30Mn-10.5Al-1.1C-(0–5) Mo steel. It was found that Mo-enriched $M_6C$ and $M_{23}C_6$ carbides precipitate during the solution treatment when the Mo content is higher than 4%, inhibiting the growth of the austenite grains during the solution treatment. Liu et al. [9] heated Fe–11Mn–10Al-1.25C steel and cooled the steel via three different methods. It was found that the distribution of κ-carbide in the samples after intense cooling was more uniform than that of the samples after air and water cooling; this, the corresponding mechanical properties were superior. However, the research reports on the hot deformation behavior of low-density high-strength Fe–Mn–Al–C steel are not thorough enough. In this paper, the microstructure evolution law and DRX behavior of the steel were studied using a thermal simulation compression experiment and metallographic observation. The flow stress curve characteristics of the steel were analyzed through the experimental data, and the dynamic recrystallization activation energy and Z parameters were calculated. The critical strain conditions for DRX were determined based on the Cingara-McQueen equation [10]. Functional relationships between rheological stress curve characteristic values (critical strain/stress, peak strain/stress and steady-state strain/stress) and the dimensionless parameter Zener-Hollomon [11] were established. A dynamic recrystallization state diagram was drawn and a dynamic recrystallization average grain size prediction model was constructed. The research results provide theoretical support for the determination of the hot forging processes parameters as well as the numerical simulation of the hot forming of low-density high-strength Fe–Mn–Al–C steel.

## 2. Materials and methods

The material used in the experiment is Fe–Mn–Al–C steel and its main chemical composition (mass fraction, %) is: 1.46 C, 25.14 Mn, 10 Al, 0.053 Nb and the rest is Fe. The corresponding initial microstructure is shown in Fig. 1. The forging billet was processed into a cylindrical sample of $\Phi$ 8 mm × 12 mm, and the perpendicularity of both ends of the samples was ensured. The isothermal constant strain rate compression experiments were carried out with the Gleeble-3500 thermo-mechanical simulator. The experimental temperatures were 900, 950, 1000, 1050, 1100 and 1150 °C, and the strain rates were 0.01, 0.1, 1 and 10 $s^{-1}$, respectively. The maximum compression was 60% (corresponding to the maximum true strain 0.92). The mica foils with high-temperature lubricants were affixed to the two end faces of the sample to reduce the friction and the non-uniformity of deformation. The sample was heated to 1200 °C at a rate of 10 °C/s, then homogenized by holding for 120 s, and finally cooled to the set $T$ at a rate of 5 °C/s. The sample was for 15 s at the set $T$ before compression, so the temperature difference between the inside and outside of the sample was the same. After compression, the sample was sprayed with water to cool the steel down. Figure 2 shows the experimental process. The compressed sample was split in half along the axial axis, and after sample preparation, grinding, polishing and etching, the microstructure images were collected using a LEXT OLS4000 laser confocal microscope. The etching solution was: 30% nitric acid alcohol solution (volume fraction). Through the Gleeble-3500 thermo-mechanical simulator, the isothermal constant strain rate thermal compression experiments of the same Fe–Mn–Al–C steel batch with the same composition can be confirmed under the same experimental conditions in order to check the reproducibility of the proposed system.

## 3. Results and discussion

### 3.1. *Rheological behavior analysis*

Fig. 3 (a) and (b) show the rheological stress curves of Fe–Mn–Al–C steel at different $T$ (900–1150 °C) and $\dot{\varepsilon}$ (0.01–10 $s^{-1}$). Fig. 3 (a) and (b) show that the rheological stress behavior of the steel is primarily influenced by $T$ and $\dot{\varepsilon}$. The decrease of $\dot{\varepsilon}$or the increase of $T$ will result in the decrease of the rheological stress, indicating that the steel is a negative temperature-sensitive and positive strain rate-sensitive material. The former is primarily due to the accelerated grain boundary mobility of the dynamic recrystallization at a higher $T$. Furthermore, high temperatures are beneficial in strengthening the softening effect and reducing the

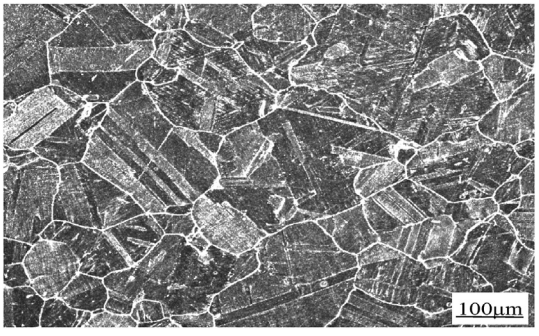



**Fig. 1 – Initial microstructure of Fe–Mn–Al–C steel.**

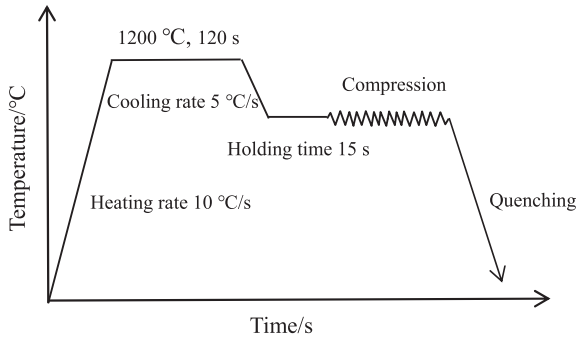



**Fig. 2 – Compression experimental process.**

dislocation slip resistance. The latter is primarily due to the lower $\dot{\varepsilon}$ providing more time to promote DRX and dynamic recovery (DRV). This leads to the work hardening effect being less than the softening effect of the material, showing that the rheological stress decreases with the decrease of $\dot{\varepsilon}$. In addition, peak stress is an important basis for estimating maximum deformation load, die selection and equipment checking in actual production. Fig. 3 (c) and (d) are 3D peak stress maps of Fe–Mn–Al–C steel under different $T$ and $\dot{\varepsilon}$. Fig. 3 (c) and (d) show that the peak stress of this steel is sensitive to $\dot{\varepsilon}$ and $T$ during hot deformation. At the low $\dot{\varepsilon}$ (0.01 $s^{-1}$), the peak stress decreased from 305.4 MPa at 900 °C to 53.8 MPa at 1150 °C, a reduction of 82%. When the $\dot{\varepsilon}$ is high (10 $s^{-1}$), the peak stress decreases by 73% within the same temperature range, indicating that $\dot{\varepsilon}$ has a greater influence on the peak stress. The main reason is that the deformation time of the material is shortened with the increase of $\dot{\varepsilon}$, the dislocation propagation rate is reduced, the DRV and DRX are insufficient, the softening effect is weak and the effect of work hardening is intensified, so the peak stress increases [12,13]. The same calculation can be obtained, in the case of a constant $\dot{\varepsilon}$, the higher the $T$, the lower the peak stress, indicating that the peak stress is more sensitive to $T$. This is because the higher the $T$, the higher the thermal activation energy of the material, the faster the grain boundary mobility, the better the recrystallization and the lower the slip resistance, the stronger the softening effect and the lower the peak stress [14,15]. In summary, the steel can reduce the peak stress by reducing $\dot{\varepsilon}$ or increasing $T$ during hot deformation. The above experimental phenomena have reproducibility.

### 3.2. *Determination of dynamic recrystallization activation energy and Z parameter*

The low-density high-strength steel underwent a greater degree of DRX during the hot deformation. The corresponding rheological stress curve has an obvious stress peak; thus, there is a peak stress/strain ($\sigma_p/\varepsilon_p$). Since the thermal deformation process of the metals was controlled by hot activation, the relationship between $\sigma_p$, $T$ and $\dot{\varepsilon}$ can be established by the Arrhenius hyperbolic sine function equation in order to determine the dynamic recrystallization activation energy:

$$\dot{\varepsilon} = A[\sinh(\alpha\sigma_p)]^n \exp\left(-\frac{Q_p}{RT}\right) \quad (1)$$

where $Q_p$ is the activation energy for dynamic recrystallization ($J \cdot mol^{-1}$); R is the gas constant ($8.314\ J \cdot (mol \cdot K)^{-1}$); $\dot{\varepsilon}$ is the

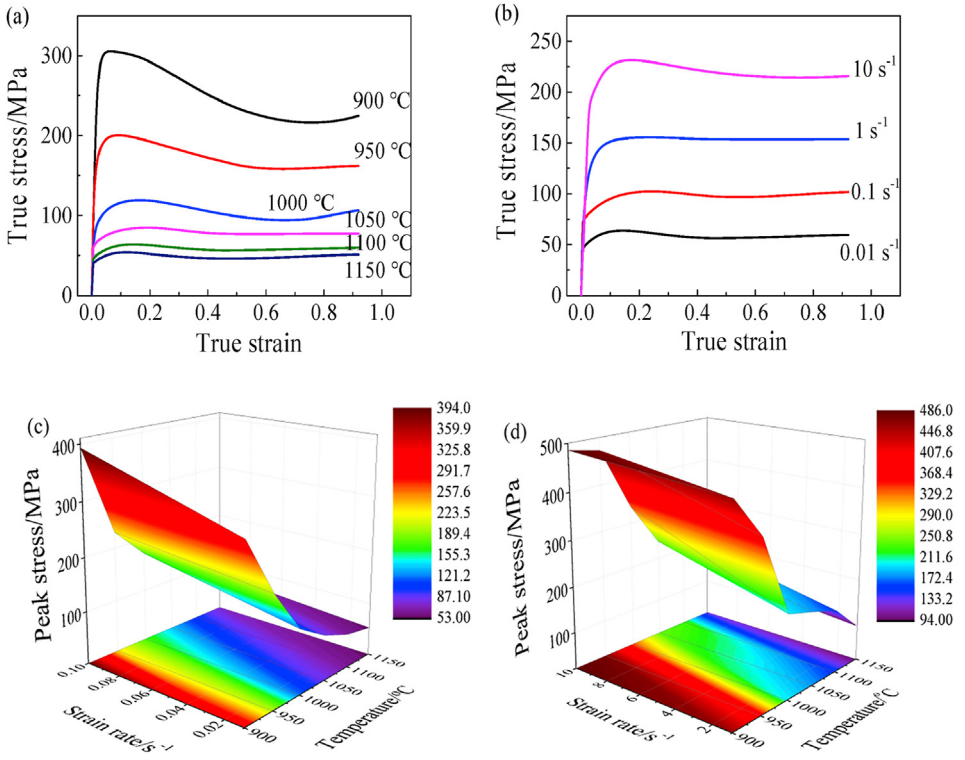



**Fig. 3 – The rheological stress curves of Fe–Mn–Al–C steel under different $T$ and $\dot{\varepsilon}$ ((a) $\dot{\varepsilon} = 0.01\ s^{-1}$; (b) $T = 1100$ °C) and the effects of different deformation conditions for the steel peak stress ((c) $\dot{\varepsilon} = 0.01–0.1\ s^{-1}$; (d) $\dot{\varepsilon} = 1–10\ s^{-1}$).**

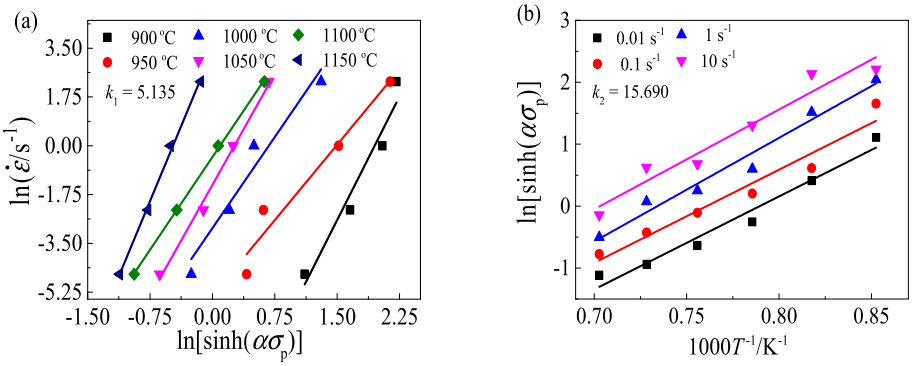



**Fig. 4 – Relationship maps of $\ln\dot{\varepsilon} - \ln[\sinh(\alpha\sigma_p)]$ (a) and $\ln[\sinh(\alpha\sigma_p)] - 1000/T$ (b).**

strain rate ($s^{-1}$); $\sigma_p$ is the peak stress (MPa); $T$ is the temperature (K); A, $\alpha$ and $n$ are all material constants, where the stress level factor $\alpha$ is determined by the two constants ($n_1$ and $\beta$) [16,17], that is $\alpha = \beta/n_1 = (\partial\ln\dot{\varepsilon}/\partial\sigma)/(\partial\ln\dot{\varepsilon}/\partial\ln\sigma) = 0.00599$.

Take the natural logarithm on both sides of formula (1), and the relationship between $\ln[\sinh(\alpha\sigma_p)] - 1000/T$ and $\ln\dot{\varepsilon} - \ln[\sinh(\alpha\sigma_p)]$ can be established, as shown in Fig. 4. According to the linear fitting, the average slopes of the curves in Fig. 4 (a) and (b) are $k_1 = 5.135$ and $k_2 = 15.690$, respectively. Furthermore, the dynamic recrystallization activation energy of Fe–Mn–Al–C steel is 669.844 kJ $mol^{-1}$ according to formula (2):

$$Q_p = 1000 \cdot R\left(\frac{\partial\ln\dot{\varepsilon}}{\partial\ln[\sinh(\alpha\sigma_p)]}\right)_T\left(\frac{\partial\ln[\sinh(\alpha\sigma_p)]}{\partial(1000/T)}\right)_{\dot{\varepsilon}} = 1000 \cdot Rk_1k_2 \tag{2}$$

The parallelism of the fitted straight line in Fig. 4 is not good, indicating that the deformation conditions have a significant effect on the dynamic recrystallization activation energy. Therefore, the calculation results of the activation energy for dynamic recrystallization above are only the average value within the experimental range of this paper. Zener and Hollomon [11] studied the constitutive relationship of steel and found that rheological stress depends on temperature and strain rate. Therefore, they proposed the Zener-Hollomon (Z) parameter model to characterize the comprehensive influence of $T$ and $\dot{\varepsilon}$ on the rheological stress.

$$Z = \dot{\varepsilon}\exp\left(Q_p/RT\right) \tag{3}$$

The temperature compensated strain rate function (Z parameter formula (3)) has been applied in many studies. Singh et al. [18] studied the dynamic recrystallization of a high Nb-containing cast γ-TiAl based Ti–45Al–8Nb–2Cr-0.2B alloy during hot compressive deformation and analyzed the Zener-Hollomon parameter modulated recrystallization mechanisms. Wang et al. [19] conducted the constitutive analysis of as-cast 40CrNiMo alloy steel through isothermal compression experiments and established the Arrhenius-type constitutive equation with the Zener-Hollomon parameter, having excellent accuracy. Detrois et al. [20] studied the hot deformation behavior of a Ni-based superalloy and simulated the flow stress from the experimental data using the Zener-Hollomon parameter.

According to the calculated dynamic recrystallization activation energy and formula (3), the Z parameters of Fe–Mn–Al–C steel under different $T$ and $\dot{\varepsilon}$ can be obtained, as shown in Table 1. Table 1 shows that the Z parameter increases with the increase of $\dot{\varepsilon}$ or the decrease of $T$, which is the same as that of the flow stress. The above experimental phenomena have reproducibility. The introduction of the Zener-Hollomon parameter helps construct the functional relationship between the characteristic value (critical strain/stress ($\varepsilon_c/\sigma_c$), peak strain/stress ($\varepsilon_p/\sigma_p$) and steady-state strain/stress ($\varepsilon_s/\sigma_s$)) of the flow stress curve and the Z parameter in the future. This also provides a practical basis for further studying the influence of hot deformation parameters (temperature and strain rate) on $\varepsilon_c/\sigma_c$, $\varepsilon_p/\sigma_p$ and $\varepsilon_s/\sigma_s$.

### 3.3. *The relationship between the characteristic value of flow stress curve and Z parameter and the dynamic recrystallization state diagram*

The flow stress curve of the DRX behavior of low-density high-strength steel can be described by three sets of characteristic values, namely critical strain/stress ($\varepsilon_c/\sigma_c$), peak strain/stress ($\varepsilon_p/\sigma_p$) and steady-state strain/stress ($\varepsilon_s/\sigma_s$), where $\varepsilon_c$ and $\varepsilon_s$ represent the strain required to initiate and complete dynamic recrystallization, respectively. The $\varepsilon_p/\sigma_p$ and $\varepsilon_s/\sigma_s$ here can be obtained directly from the rheological stress curve, while $\varepsilon_c/\sigma_c$ can be obtained only through data processing. In this regard, the $\varepsilon_c$ of DRX was determined by the mathematical derivation of the Cingara-McQueen equation. The formula is shown in (4):

**Table 1 – lnZ value of Fe–Mn–Al–C steel under different $T$ and $\dot{\varepsilon}$.**

| $\dot{\varepsilon}$ ($s^{-1}$) | T (°C) | | | | | |
|---|---|---|---|---|---|---|
| | 900 | 950 | 1000 | 1050 | 1100 | 1150 |
| 0.01 | 64.08 | 61.27 | 58.69 | 56.29 | 54.08 | 52.01 |
| 0.1 | 66.38 | 63.58 | 60.99 | 58.60 | 56.38 | 54.32 |
| 1 | 68.69 | 65.88 | 63.29 | 60.90 | 58.68 | 56.62 |
| 10 | 70.99 | 68.18 | 65.59 | 63.20 | 60.98 | 58.92 |

$$\frac{\sigma}{\sigma_p}=\left[\frac{\varepsilon}{\varepsilon_p}\exp\left(1-\frac{\varepsilon}{\varepsilon_p}\right)\right]^C \tag{4}$$

where C is the material constant. Take the natural logarithm on both sides of formula (4), which can be written into formula (5):

$$\ln\left(\frac{\sigma}{\sigma_p}\right)=C\left[\ln\left(\frac{\varepsilon}{\varepsilon_p}\right)+1-\frac{\varepsilon}{\varepsilon_p}\right] \tag{5}$$

The formula (5) shows that C can be calculated from the slope fitted by $\ln(\sigma/\sigma_p)-[\ln(\varepsilon/\varepsilon_p)+1-\varepsilon/\varepsilon_p]$. The fitting results at 1050 °C and 0.01 $s^{-1}$ are shown in Fig. 5. According to this method, the material constant C under other T and $\dot{\varepsilon}$ can also be obtained.

The work hardening rate $\theta$ expression can be derived by the partial differentiation of formula (4), that is:

$$\theta=\partial\sigma/\partial\varepsilon=C\sigma(1/\varepsilon-1/\varepsilon_p) \tag{6}$$

Poliak and Jonas [21] concluded that when dynamic recrystallization occurred, the $\theta-\sigma$ curve would present a characteristic inflection point, that is, $-\partial^2\theta/\partial\sigma^2=0$, the inflection point was the critical point of dynamic recrystallization. Using the properties of partial derivatives to formula (6), the following relationship can be derived:

$$\partial\theta/\partial\sigma=\partial\ln\theta/\partial\varepsilon \tag{7}$$

It is seen that there must be an inflection point corresponding to the $\theta-\sigma$ curve in the $\ln\theta-\varepsilon$ curve, that is, $-\partial^2\ln\theta/\partial\varepsilon^2=0$. In this regard, by taking the natural logarithm of formula (7), and performing the second partial derivative:

$$\frac{\partial}{\partial\varepsilon}\left(-\frac{\partial\ln\theta}{\partial\varepsilon}\right)=0 \tag{8}$$

The mathematical expression of the $\varepsilon_c$ of DRX can be obtained:

$$\frac{\varepsilon_c}{\varepsilon_p}=\frac{\sqrt{1-C}-(1-C)}{C} \tag{9}$$

According to the C values obtained under different T and $\dot{\varepsilon}$, the corresponding $\varepsilon_c$ can be calculated via formula (9) and the $\sigma_c$ corresponding to the $\varepsilon_c$ can be calculated based on the flow stress curve. The functional relationship between the required $\varepsilon_c$ and the $\varepsilon_p$ is $\varepsilon_c/\varepsilon_p\approx 0.452$. This result is similar to the conclusions found within the literature [22,23] that use the inflection point on the $\theta-\sigma$ curve to identify the dynamic recrystallization $\varepsilon_c$ of a low-density high-strength steel, indicating that the method is reliable in solving the critical strain of the steel.

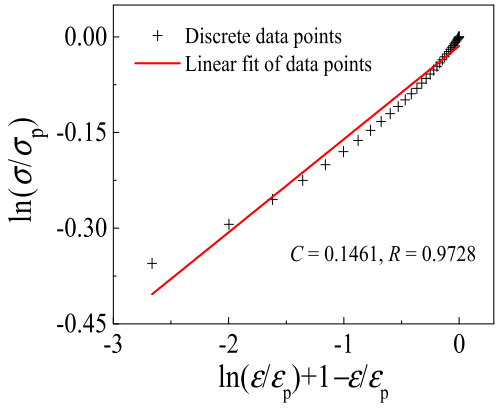



**Fig. 5 – Relationship of $\ln(\sigma/\sigma_p)-[\ln(\varepsilon/\varepsilon_p)+1-\varepsilon/\varepsilon_p]$ at 1050 °C, 0.01 $s^{-1}$.**

The Z parameter in Table 1 is introduced to characterize the functional relationship between the characteristic values of the rheological stress curves of Fe–Mn–Al–C steel, as shown in Fig. 6. The mathematical models of $\varepsilon_c/\sigma_c$, $\varepsilon_p/\sigma_p$ and $\varepsilon_s/\sigma_s$ of the dynamic recrystallization are obtained by multiple linear regression fittings:

$$\begin{cases}\varepsilon_c=0.0093\ln Z-0.43\\ \sigma_c=1.0048(\ln Z)^2-97.23\ln Z+2379.08\\ \varepsilon_p=0.020\ln Z-0.94\\ \sigma_p=0.98(\ln Z)^2-94.21\ln Z+2291.34\\ \varepsilon_s=0.034\ln Z-1.53\\ \sigma_s=0.55(\ln Z)^2-46.96\ln Z+984.67\end{cases} \tag{10}$$

According to the relationship between $\varepsilon_c$, $\varepsilon_s$ and the Z parameter above, the dynamic recrystallization state diagram of hot deformation of Fe–Mn–Al–C steel can be drawn, as shown in Fig. 7, in which area A is the zone of work hardening, area B is the zone of partial DRX and area C is the zone of complete DRX. As it can be seen from Fig. 7, with the increase of the Z parameter, that is, with the increase of $\dot{\varepsilon}$ or the decrease of T, the $\varepsilon_c$ and $\varepsilon_s$ of dynamic recrystallization increases. It indicates that the behavior of starting DRX and completing DRX of the steel has become more difficult. The above experimental phenomena have reproducibility.

### 3.4. *Microstructure evolution*

During hot deformation, the microstructure of DRX directly affects the grain size of the forgings and then affects the mechanical properties of the materials. Due to the friction between the contact surface and the pressure head during compression, the stress in the material is unevenly distributed, leading to different deformation zones in the section parallel to the loading direction of the compression sample, as shown in Fig. 8. Among them, 1 is a difficult deformation zone, 2 is a small deformation zone and 3 is a large deformation zone. To comprehensively analyze the evolution law of microstructure of Fe–Mn–Al–C steel during hot deformation, the microstructure of three deformation zones in Fig. 8 was observed. The results are shown in Fig. 9 (1, 2 and 3 in Fig. 9 represent the corresponding microstructure of the zone of difficult deformation, the zone of small deformation and the zone of large deformation, respectively).

Figure 9 (a)-(c) shows the microstructure morphology of Fe–Mn–Al–C steel in different deformation zones after 60% compression at 1100 °C under different $\dot{\varepsilon}$. As it can be seen from Fig. 9 (c), the grain boundaries of the large deformation zone were serrated at $\dot{\varepsilon}=10\ s^{-1}$. The DRX grains are distributed along the elongated original grain boundary in the form of a small and equiaxed necklace. With the decrease of $\dot{\varepsilon}$, the recrystallized grains grew and the degree of microstructure heterogeneity improved (Fig. 9 (a) and (b)). Under the same deformation conditions, there were obvious differences in the

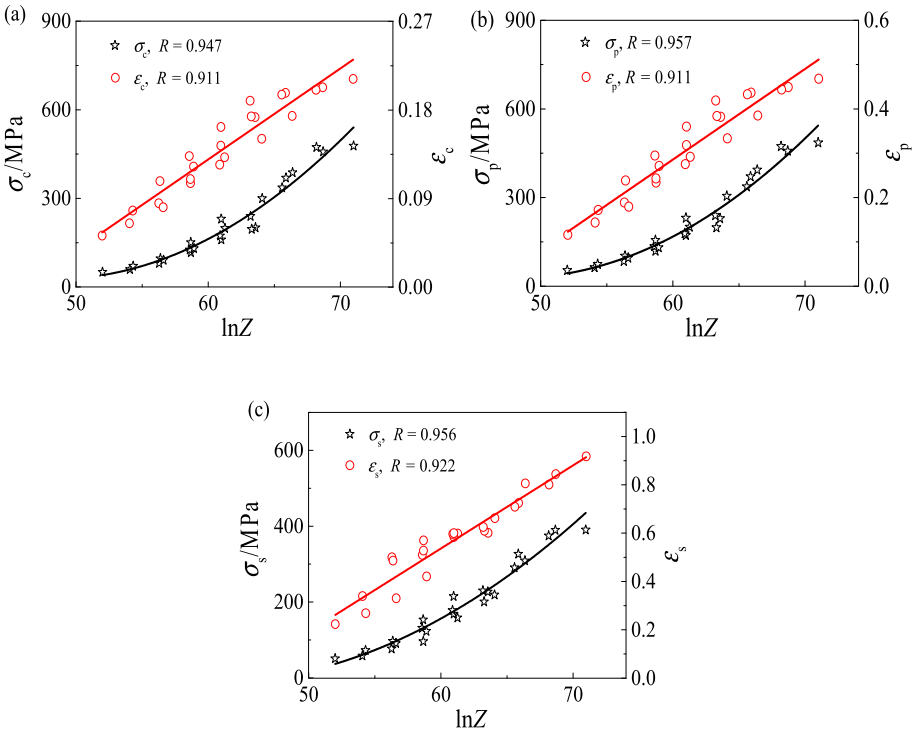



**Fig. 6 – Relationship between characteristic values of rheological stress curves and lnZ for the steel: (a) $\sigma_c, \varepsilon_c$ – ln Z; (b) $\sigma_p, \varepsilon_p$ – ln Z; (c) $\sigma_s, \varepsilon_s$ – ln Z.**

microstructure of the three deformation zones. The degree of dynamic recrystallization was zone 3 of large deformation > zone 2 of small deformation > zone 1 of difficult deformation. Figure 9 (d)-(f) shows the microstructures of the steel at different $T$ when the $\dot{\varepsilon}$ was 1 $s^{-1}$. It was found that $T$ has a significant effect on the DRX structure. With the increase of $T$, the volume fraction of DRX increases. At 1050 °C, the dynamic recrystallization is basically completed within the large deformation zone, and the recrystallized grains size grows significantly with the increase of $T$. This may be because the dislocation motion is a hot activation process. The increase of $T$ will increase the oscillation frequency of the atoms, promote the diffusion of atoms, the grain boundary migration and the dislocation motion, and then effectively induce the occurrence of DRX and make it fully performed [24,25]. Figure 10 shows the dynamic recrystallization average grain size of the steel under different $T$ and $\dot{\varepsilon}$ when the hot compression reaches 60%. The microstructure of the measured grain size is taken from the core of the compressed sample, which is the large deformation zone. Figure 10 shows that the DRX grain size of the steel after hot deformation increases with the decrease of $\dot{\varepsilon}$ or the increase of $T$. Among them, when the $T$ is 1150 °C and the $\dot{\varepsilon}$ is 0.01 $s^{-1}$, the grain size exceeds 70 μm, and the structure is coarse grain, indicating that the recrystallized grains under this condition have grown rapidly. However, at low $T$ (900–950 °C) and high $\dot{\varepsilon}$ (1–10 $s^{-1}$), the dynamic recrystallization grains are small, but not sufficient and the original grains with large size remained (Fig. 9 (d)), and the microstructure homogeneity was poor.

In conclusion, fine dynamic recrystallization grains tend to occur at low $T$ and high $\dot{\varepsilon}$ regions, but are not sufficient. The dynamic recrystallization average grain size obtained at high $T$ and low $\dot{\varepsilon}$ is large, which does not meet the fine grain size requirements. Therefore, combined with the observation of

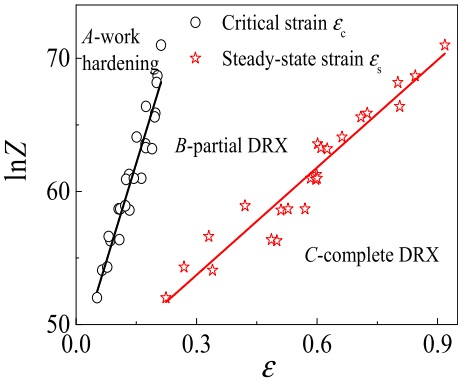



**Fig. 7 – Dynamic recrystallization state diagram for the steel under hot deformation.**

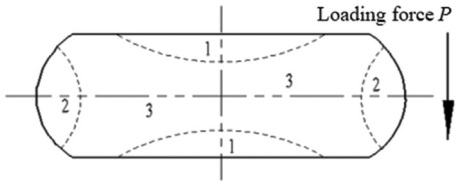



**Fig. 8 – Different deformation zones for the sample after compression: 1-difficult deformation zone; 2-small deformation zone; 3-large deformation zone.**

the microstructure and the variation of the dynamic recrystallization average grain size, sufficient dynamic recrystallization can be obtained by hot working within the range of 950–1000 °C/0.01–0.1 $s^{-1}$ and 1050–1100 °C/1–10 $s^{-1}$, and the microstructure of the grain is relatively fine and uniform. The above experimental phenomena have reproducibility.

### 3.5. *Dynamic recrystallization average grain size prediction model*

To effectively provide a prediction model for the dynamic recrystallization average grain size for the numerical simulation software, the finite element software DEFORM-3D model was adopted:

$$d_{rex} = a d_0^{\ h} \varepsilon^p \dot{\varepsilon}^q \exp(Q/RT) + b \quad (11)$$

where $a$, $h$, $p$, $q$ and $b$ are all parameters to be determined; $d_0$ is the initial grain size. The formula (11) shows that the primary factors affecting the average grain size of dynamic recrystallization are the initial grain size, $T$, $\dot{\varepsilon}$ and strain. In this experiment, the initial grain size and strain of the material can be taken as fixed values, that is, $a d_0^{\ h} \varepsilon^p$ can be regarded as a constant. Under high temperature conditions, the plastic deformation of the workpiece was relatively large, and the elastic deformation was usually negligible. Therefore, in the finite element simulation, the billet was set as a rigid plastic body and the indenter of the die was set as a rigid body. In order to improve the accuracy of the simulation, the material parameters were given to the die. A semi-symmetric specimen was used as the deformation model to shorten the finite element simulation time. Among them, it should be noted that the friction of the contact interface between the billet and the die was set to the shear type, and the value was 0.3 [26,27]. The convection coefficient of the billet and the air was set as 0.02 N/(s·mm·°C). The simulation was in line with the experimental conditions of thermal compression. The

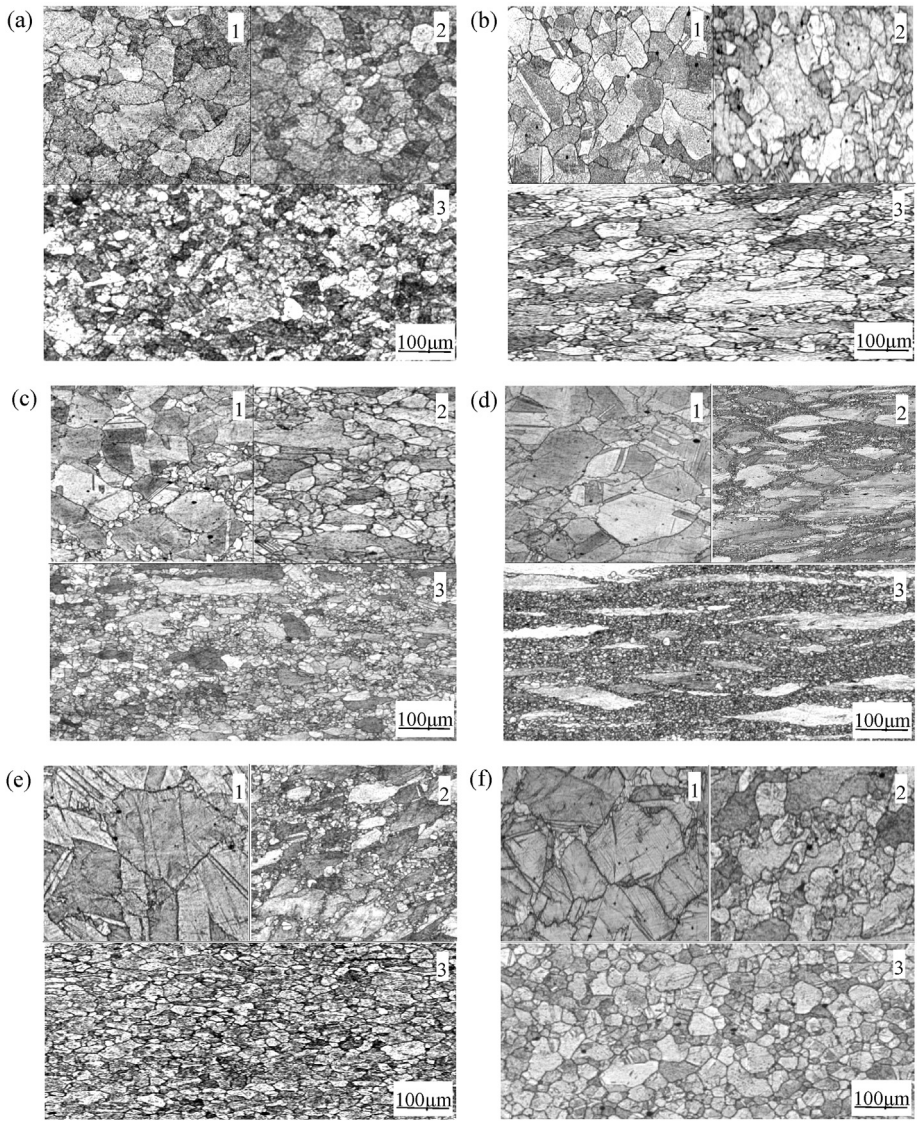



**Fig. 9 – Microstructure of Fe–Mn–Al–C steel under different $T$ and $\dot{\varepsilon}$: (a) $T = 1100$ °C, $\dot{\varepsilon} = 0.01$ $s^{-1}$; (b) $T = 1100$ °C, $\dot{\varepsilon} = 0.1$ $s^{-1}$; (c) $T = 1100$ °C, $\dot{\varepsilon} = 10$ $s^{-1}$; (d) $T = 950$ °C, $\dot{\varepsilon} = 1$ $s^{-1}$; (e) $T = 1050$ °C, $\dot{\varepsilon} = 1$ $s^{-1}$; (f) $T = 1150$ °C, $\dot{\varepsilon} = 1$ $s^{-1}$.**

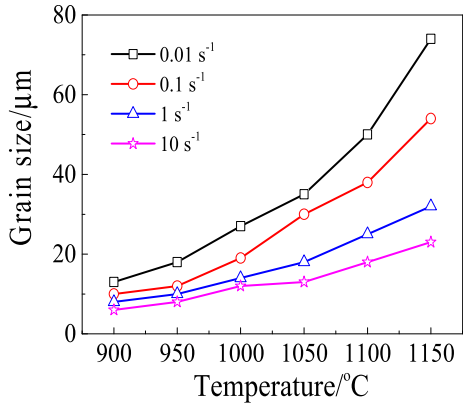



**Fig. 10 – Dynamic recrystallization average grain size of the steel under different T and $\dot{\varepsilon}$.**

Levenberg–Marquardt optimization algorithm was used to perform the mathematical regression analysis on the experimental data, and the prediction model for the average grain size of the dynamic recrystallization of the low-density high-strength steel was obtained:

$$d_{rex} = 807743.63\dot{\varepsilon}^{-0.1903} \exp(-120648.47/RT) + 4.338 \quad (12)$$

In order to verify the accuracy of the prediction model, the correlation analysis and error test of the predicted value of the grain size and the experimental value were carried out. The accuracy of the chosen model can be assessed through the following two parameters: *R* and *MRE*. They can be defined as:

$$R = \frac{\sum_{i=1}^{n}\left(M_i - \overline{M}\right)\left(N_i - \overline{N}\right)}{\sqrt{\sum_{i=1}^{n}\left(M_i - \overline{M}\right)^2}\sqrt{\sum_{i=1}^{n}\left(N_i - \overline{N}\right)^2}} \quad (13)$$

$$MRE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{M_i - N_i}{M_i}\right| \times 100\% \quad (14)$$

where $M_i$ and $N_i$ represent the experiment data of the testing and the corresponding predicted data acquired from the model; $n$ is the number of testing data; $\overline{M}$ and $\overline{N}$ are the average values of $M_i$ and $N_i$, respectively. The correlation coefficient (*R*) was usually used to show the linear relations between the predicted and the experimental data. To evaluate the selected model, the average relative error (*MRE*) was considered an unbiased statistic for confirming the credibility of the model.

During the dynamic recrystallization of the material, the grain size constantly changes under the driving force. On the one hand, the grain growth is promoted under the action of temperature and plastic strain and the driving force has a positive effect on the grain size. On the other hand, dynamic recrystallization refines the microstructure inside the material and the driving force is a counter-driving force, which inhibits the grain growth [27–29]. This may cause errors in the dynamic recrystallization average grain size prediction model. In addition, there were some statistical errors. The results are shown in Fig. 11, *R* is 0.996 and *MRE* for all samples is only 4.53%, indicating a high degree of coincidence. Figure 11 (c) shows that the absolute error between the predicted value of

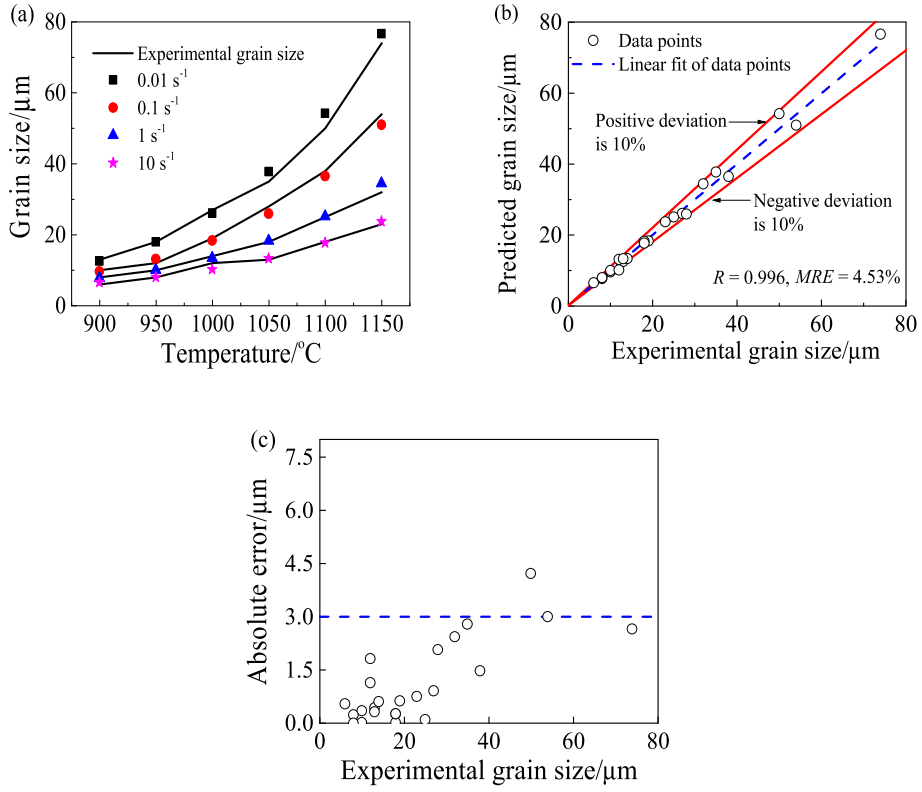



**Fig. 11 – The correlation and absolute error between predicted and experimental values of dynamic recrystallization grain size model.**

the grain size and the experimental value is mostly within 3 μm, which further shows that the model can accurately predict the dynamic recrystallization average grain size of low-density high-strength steel. It provides a reliable theoretical basis for the prediction of the dynamic recrystallized grain size of the steel after hot forging.

## 4. Conclusion and prospect

### 4.1. Conclusion

(1) Fe–Mn–Al–C steel is a negative temperature-sensitive and positive strain rate-sensitive material, and its rheological stress increases with the increase of $\dot{\varepsilon}$ or the decrease of $T$. During hot deformation, the peak stress of the steel can be reduced by decreasing $\dot{\varepsilon}$ or increasing $T$.

(2) Based on the Arrhenius hyperbolic sine function equation, the dynamic recrystallization activation energy of the steel was calculated as 669.844 kJ $mol^{-1}$. Through the introduction of the dimensionless parameter Zener-Hollomon, the $\varepsilon_c/\sigma_c$, $\varepsilon_p/\sigma_p$ and $\varepsilon_s/\sigma_s$ models of dynamic recrystallization were established, and the state diagram of DRX was drawn. The results show that with the increase of the $Z$ parameter, the initiation and completion of the dynamic recrystallization behavior of the steel becomes more difficult.

(3) The low $T$ and high $\dot{\varepsilon}$ are advantageous to obtain fine dynamic recrystallization grains, but they are not sufficient. With the decrease of $\dot{\varepsilon}$ and the increase of $T$, full dynamic recrystallization of the material can be effectively induced and grain growth can also be caused. It is considered by comprehensive analysis that the fine and uniform grain structure can be obtained by hot working in the range of 950–1000 °C/0.01–0.1 $s^{-1}$ and 1050–1100 °C/1–10 $s^{-1}$.

(4) The prediction model of dynamic recrystallization average grain size is $drex = 807743.63\dot{\varepsilon}^{-0.1903}\exp(-120648.47/RT) + 4.338$, and the deviation is generally lower than 3 μm, and $MRE$ for all samples is only 4.53%, which indicates that the model can be used to predict the grain size of low-density high-strength steel during hot deformation.

### 4.2. Prospect

When setting the experimental conditions, we considered different parameters and obtained the best value of each parameter. Using multivariate research (experimental design and chemometrics), optimizing of two or more parameters is easier, more statistically correct and functional. At present, this multivariate method (experimental design) has been used in other studies (such as energy, environment and sensors). Hendriksen et al. [30] expanded the application of Noise2-Inverse in space, time and spectrum-like domains based on convolutional neural networks. The results on real-world datasets establish that Noise2Inverse can accurately denoise enabling a substantial reduction in the acquisition time while maintaining image quality. Galliano et al. [31] used the experimental design method to accurately study the effects of $Co^{2+}$ concentration, $Co^{2+}/Co^{3+}$ ratio and xanthan gum dosage on the overall photovoltaic parameters of laboratory-scale solar cells at a multivariate level. Miccoli et al. [32] presented an efficient multisensor based on a single zinc oxide microwire/gold junction. The device can detect in real time three different stimuli, UV-VIS light, temperature and pH variations. It shows great potential for use in the field of biosensors. Yun et al. [33] used headspace vaporized GC/MS (HS-GC/MS) to analyze black tea samples collected from 10 geographic origins. The findings support the use of HS-GC/MS combined with chemometrics as a tool to identify the origin of black tea. Galliano et al. [34] have adopted an experimental strategy of multivariate chemometric approach to study the main factors influencing dye-sensitized solar cells efficiency and stability and obtained an optimized system which can improve their efficiency. Bella et al. [35] proposed the replacement of the liquid electrolyte with quasi-linear polymer electrolyte membranes, with the aim of increasing the durability of the device. The experimental conditions are investigated and optimized by means of a multivariate chemometric approach, and the characterization of the materials and devices was presented.

In conclusion, we will carry out the multivariate study in future work in order to take into account all the relevant variables and their interactions, finding the best experimental conditions for the proper functioning of their materials/devices/reactors.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## REFERENCES

[1] Kalashnikov IS, Acselrad O, Pereira LC, Kalichak T, Khadyyev MS. Behavior of Fe-Mn-Al-C steels during cyclic tests. J Mater Eng Perform 2000;9(3):334–7.

[2] Frommeyer G, Brüx U. Microstructures and mechanical properties of high-strength Fe-Mn-Al-C light-weight TRIPLEX steels. Steel Res Int 2006;77(9–10):627–33.

[3] Zuazo I, Hallstedt B, Lindahl B, Selleby M, Soler M, Etienne A, et al. Low-density steels: complex metallurgy for automotive applications. JOM 2014;66(9):1747–58.

[4] Chen SP, Rana R, Haldar A, Ray RK. Current state of Fe-Mn-Al-C low density steels. Prog Mater Sci 2017;89:345–91.

[5] Chen XF, Tang B, Liu Y, Xue XY, Li L, Kou HC, et al. Dynamic recrystallization behavior of the Ti-48Al-2Cr-2Nb alloy

during isothermal hot deformation. Prog Nat Sci: Mater Int 2019;29:587–94.
[6] Abbasi SM, Momeni A, Lin YC, Jafarian HR. Dynamic softening mechanism in Ti-13V-11Cr-3Al beta Ti alloy during hot compressive deformation. Mater Sci Eng A 2016;665:154–60.
[7] Ji HC, Duan HL, Li YG, Li WD, Huang XM, Pei WC, et al. Optimization the working parameters of as-forged 42CrMo steel by constitutive equation-dynamic recrystallization equation and processing maps. J Mater Res Technol 2020;9(4):7210–24.
[8] Moon J, Park SJ, Jang JH, Lee TH, Lee CH, Hong HU, et al. Investigations of the microstructure evolution and tensile deformation behavior of austenitic Fe-Mn-Al-C lightweight steels and the effect of Mo addition. Acta Mater 2018;147:226–35.
[9] Liu DG, Cai MH, Ding H, Han D. Control of inter/intra-granular κ-carbides and its influence on overall mechanical properties of a Fe-11Mn-10Al-1.25C low density steel. Mater Sci Eng A 2018;715:25–32.
[10] Cingara A, Mcqueen HJ. New method for determining sinh constitutive constants for high temperature deformation of 300 austenitic steels. J Mater Process Technol 1992;36(1):17–30.
[11] Zener C, Hollomon JH. Effect of strain rate upon plastic flow of steel. J Appl Phys 1944;15(1):22–32.
[12] Zhao HT, Liu GQ, Xu L. Rate-controlling mechanisms of hot deformation in a medium carbon vanadium microalloy steel. Mater Sci Eng A 2013;559:262–7.
[13] Tan YB, Ma YH, Zhao F. Hot deformation behavior and constitutive modeling of fine grained Inconel 718 superalloy. J Alloys Compd 2018;741:85–96.
[14] Wan P, Zou H, Wang KL, Zhao ZZ, Lu SQ, Nie HF. Hot deformation behaviors and intrinsic hot workability map of Ti-12Mo-4Zr-5Sn alloy based on physical model and polar reciprocity model. Metals 2020;10:956.
[15] Sun Y, Wan ZP, Hu LX, Ren JS. Characterization of hot processing parameters of powder metallurgy TiAl-based alloy based on the activation energy map and processing map. Mater Des 2015;86:922–32.
[16] Jonas JJ, Sellars CM, Tegart WJM. Strength and structure under hot-working conditions. Metall Rev 1969;14(1):1–24.
[17] Lin YC, Chen MS, Zhong J. Constitutive modeling for elevated temperature flow behavior of 42CrMo steel. Comput Mater Sci 2008;42(3):470–7.
[18] Singh V, Mondal C, Sarkar R, Bhattacharjee PP, Ghosal P. Dynamic recrystallization of a β(B2)-stabilized γ-TiAl based Ti-45Al-8Nb-2Cr-0.2B alloy: the contributions of constituent phases and Zener-Hollomon parameter modulated recrystallization mechanisms. J Alloys Compd 2020;828:154386.
[19] Wang W, Ma R, Li LP, Zhai RX, Ma SB, Yan HJ, et al. Constitutive analysis and dynamic recrystallization behavior of as-cast 40CrNiMo alloy steel during isothermal compression. J Mater Res Technol 2020;9(2):1929–40.
[20] Detrois M, Antonov S, Tin S, Jablonski PD, Hawk JA. Hot deformation behavior and flow stress modeling of a Ni-based superalloy. Mater Char 2019;157:109915.
[21] Poliak EI, Jonas JJ. A one-parameter approach to determining the critical conditions for the initiation of dynamic recrystallization. Acta Mater 1996;44(1):127–36.
[22] Gwon H, Shin S, Jeon J, Song T, Kim S, Cooman BC. Hot deformation behavior of V micro-alloyed TWIP steel during hot compression. Met Mater Int 2019;25:594–605.
[23] Yang FQ. Research on the preparation technology and deformation mechanism of automobile light-weight Fe-Mn-Al-C high strength steel. University of Science and Technology Beijing; 2015.
[24] Kim WJ, Jeong HT. Construction of processing maps combined with deformation mechanism maps using creep deformation equations. J Mater Res Technol 2020;9(6):13434–49.
[25] Huo YM, He T, Chen SS, Ji HC, Wu RM. Microstructure evolution and unified constitutive equations for the elevated temperature deformation of SAE 52100 bearing steel. J Manuf Process 2019;44:113–24.
[26] Liu J, Wang P. DEFORM-3D simulation of forging of semi-solid 6061 alloy and validation. Rare Met Mater Eng 2014;43(10):2455–8.
[27] Feng R, Wang KL, Lu SQ, Li X, Ouyang DL, Cui X, et al. Dynamic recrystallization behavior and numerical simulation of β phase of BT25 titanium alloy. Rare Met Mater Eng 2021;50(3):894–901.
[28] Chen MS, Yuan WQ, Lin YC, Li HB, Zou ZH. Modeling and simulation of dynamic recrystallization behavior for 42CrMo steel by an extended cellular automaton method. Vacuum 2017;146:142–51.
[29] Babu KA, Prithiv TS, Gupta A, Mandal S. Modeling and simulation of dynamic recrystallization in super austenitic stainless steel employing combined cellular automaton, artificial neural network and finite element method. Comput Mater Sci 2021;195:110482.
[30] Hendriksen AA, Bührer M, Leone L, Merlini M, Vigano N, Pelt DM, et al. Deep denoising for multi-dimensional synchrotron X-ray tomography without high-quality reference data. Sci Rep 2021;11:11895.
[31] Galliano S, Bella F, Bonomo M, Giordano F, Gratzel M, Viscardi G, et al. Xanthan-based hydrogel for stable and efficient quasi-solid truly aqueous dye-sensitized solar cell with cobalt mediator. Solar RRL 2021;5(7):2000823.
[32] Miccoli B, Cauda V, Bonanno A, Sanginario A, Bejtka K, Bella F, et al. One-dimensional ZnO/Gold junction for simultaneous and versatile multisensing measurements. Sci Rep 2016;6:29763.
[33] Yun J, Cui CJ, Zhang SH, Zhu JJ, Peng CY, Cai HM, et al. Use of headspace GC/MS combined with chemometric analysis to identify the geographic origins of black tea. Food Chem 2021;360:130033.
[34] Galliano S, Bella F, Bonomo M, Viscardi G, Gerbaldi C, Boschloo G, et al. Hydrogel electrolytes based on xanthan gum: green route towards stable dye-sensitized solar cells. Nanomaterials 2020;10:1585.
[35] Bella F, Imperiyka M, Ahmad A. Photochemically produced quasi-linear copolymers for stable and efficient electrolytes in dye-sensitized solar cells. J Photochem Photobiol Chem A 2014;289:73–80.